Satwika GV

ee24btech11027@iith.ac.in | +91 8328403988 | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.51
Sri Chaitanya Jr. college, Vijayawada	Class XII (Board of Intermediate Education,Andhra pradesh)	2024	97.90%
Balaji High School	Class X (Board of Secondary Education,Andhra Pradesh)	2022	92.67%

SKILLS

- **Languages:** C, C++, Python, SQL
- **Frameworks**: FastAPI
- **Databases**: PostgreSQL
- **Core CS**: Data Structures, Algorithms, DBMS, OOP
- **Tools**: Git, GitHub, Linux, LaTeX

RELEVANT COURSES

- **Computer science and programming** : Introduction to programming (C), Data structures and Applications,DBMS I
- **Mathematics and statistics** : Probability,Random Process, Maths for EE(linear algebra),series of function,Introduction to statistics,Numerical Analysis
- **Artificial Intelligence** : Machine Learning

PROJECTS

Huffman File Compressor:

- Built a lossless file compression and decompression utility in C++ using the Huffman Coding algorithm, supporting arbitrary binary files (e.g., text and images) with verified round-trip correctness.
- Designed a custom binary file format incorporating magic bytes, frequency table, original file size, and encoded bitstream to enable reliable decompression.
- Benchmarked compression performance against gzip on repetitive, random, and edge-case inputs (e.g., empty and single-byte files), analyzing compression ratios
- Github : https://github.com/Satwikagv/File-Compressor

Student management API:

- Designed a normalized (3NF) PostgreSQL database for a Student Management System with 6 relational tables, enforcing data integrity through primary/foreign keys
- Built a RESTful backend API using FastAPI and SQLAlchemy ORM, implementing full CRUD operations across Students, Courses, Enrollments, and Attendance modules.
- Seeded and tested the system with a realistic dataset (50 students, 150 enrollments, 900+ attendance records), validating API functionality via Swagger UI.
- Github:https://github.com/Satwikagv/student-management-api

ML algorithms from scratch:

- Implemented 10+ classical ML algorithms from scratch in NumPy (linear/Bayesian regression, LDA, Perceptron, logistic regression via IRLS, Gaussian Mixture Models with EM, KNN, and a feedforward neural network with manual backpropagation) -- no scikit-learn or autograd used for modeling.

- Built and trained a 3-layer feedforward neural network from scratch (forward pass, cross-entropy loss, and hand-derived backpropagation), achieving ~93-95% test accuracy on MNIST without any deep learning framework
- Github : https://github.com/Satwikagv/ML-algorithms

EXTRACURRICULAR

- Achieved a peak Chess.com rapid rating of **1171**.